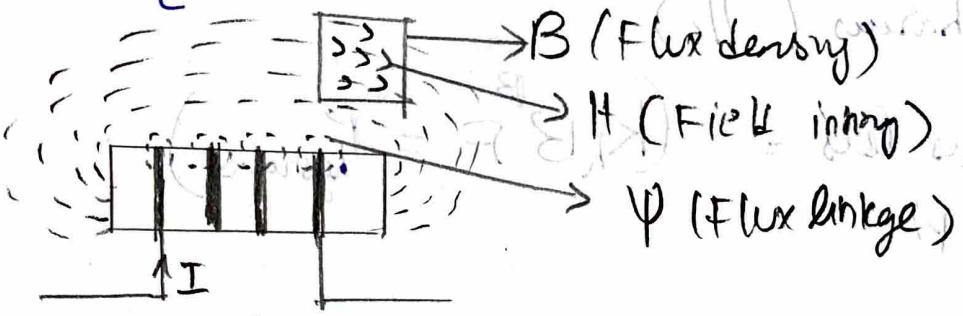
① Faradays law: $emf = -N\frac{d\phi}{dt}$; $\oint_c E \cdot dl = -\int_s \frac{dB}{dt} ds$

② Ampers law: $\oint_c H \cdot dl = I$.

③



④ $L = \frac{N\phi}{I} = \frac{\psi}{I}$

⑤ Energy stored in inductor

$$W = \frac{1}{2}\int BH\, dV$$

$$W_{total} = \frac{1}{2} \times L \times I^2$$

$$W_{core} = \frac{1}{2} \times \frac{(B_{core})^2}{\mu_{core}} \times (A_{core} \times l_{core})$$

$$\mu_{core} = \frac{B_{core}}{H_{core}}$$

$$W_{gap} = \frac{1}{2} \times \frac{(B_{gap})^2}{\mu_0} \times (A_{gap} \times g)$$

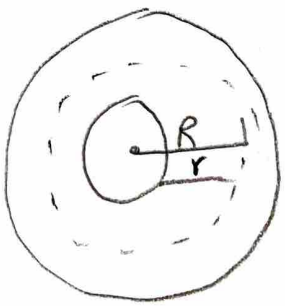
$$\mu_0 = 4\pi \times 10^{-7}$$

⑥ Solenoid core:

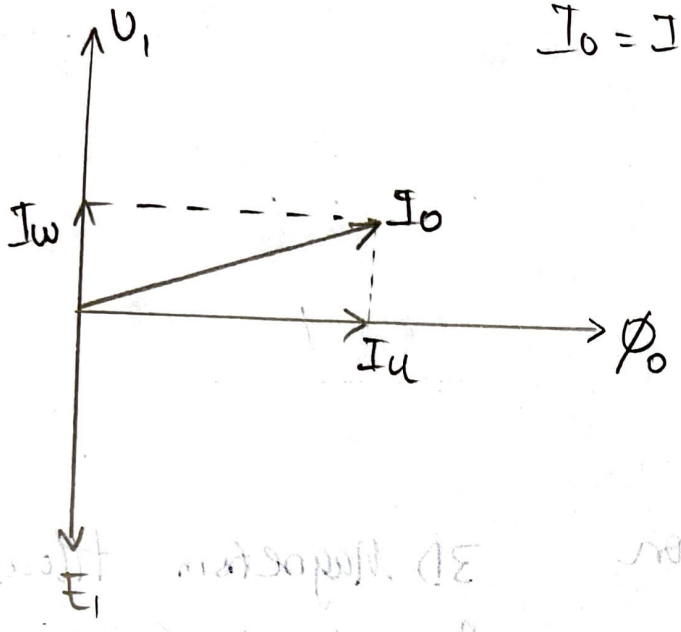
$$\text{Inductance} = L = \frac{N^2}{R_m} = \frac{N^2}{\frac{l_{core}}{\mu_0 \mu_r A_{core}} + \frac{g}{\mu_0 \mu_r A_{gap}}}$$

⑦ Toroidal core

$$L = \frac{N^2 r^2 \mu_0}{2R}$$



② Ideal Transformer [Core loss included]

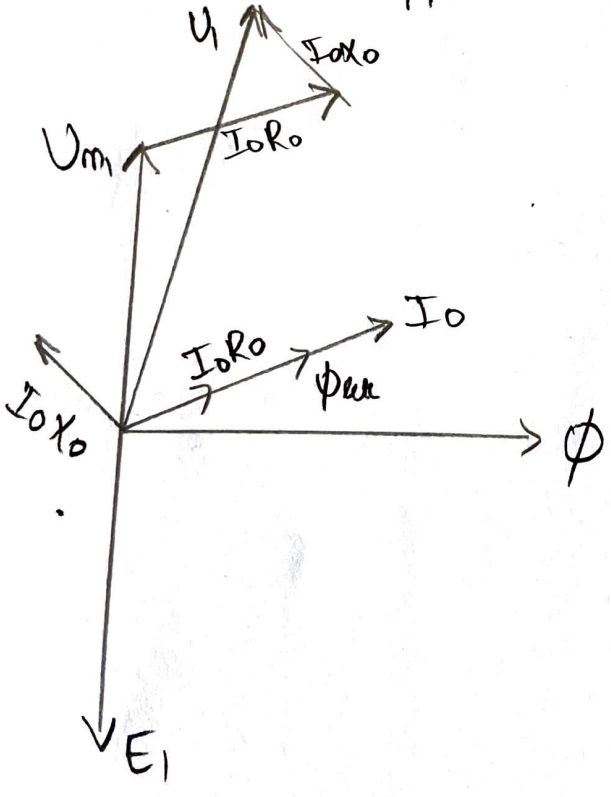


$I_o = I_\mu + I_w$

$I_\mu \rightarrow$ magnetizing component

$I_w \rightarrow$ Core loss component

3) Transformer at no loaded condition ($I\phi$) with Core and (Ideal) Copper loss



$I_o X_o \rightarrow$ V drop due to Reactance

$I_o R_o \rightarrow$ V drop due to Resistance

① Saturation Flux density (B):

Design equation $\phi = BA$; $E = 4.44 f \phi N B_m A$.

② Coercivity (Hc)

↳ hysteresis loss (loop) = $(K_h B^{\beta} f = P_{hysteresis})$

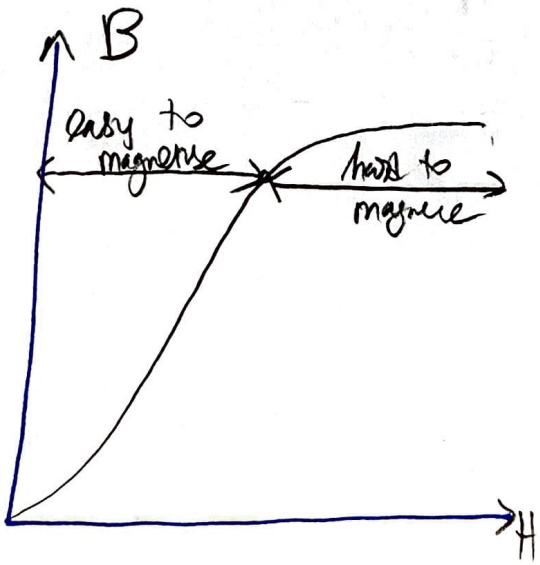
③ Resistance (R)

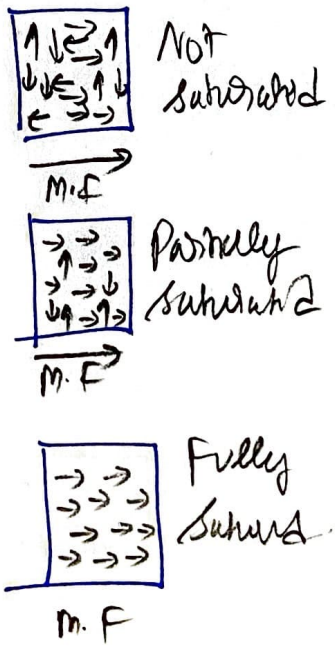
↳ Eddy current loss: Eddy current $\propto \frac{1}{\rho}$ and $\delta = \sqrt{\frac{\rho}{\pi \mu_0 \mu_r f}}$

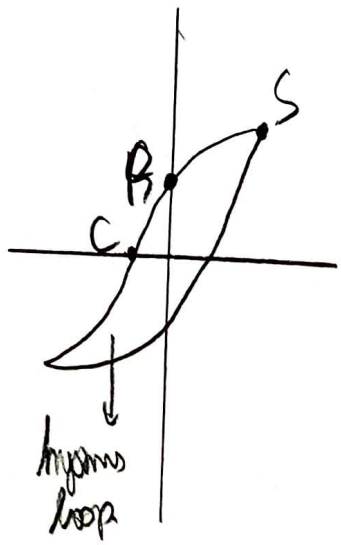
④ Permeability of the material:

$$\left[L = \frac{N^2}{R_m} = \frac{N^2}{\frac{l}{\mu_0 \mu_r A}} = \frac{N^2 \mu_0 \mu_r A}{l} \right]$$

⑤ Saturation graph:

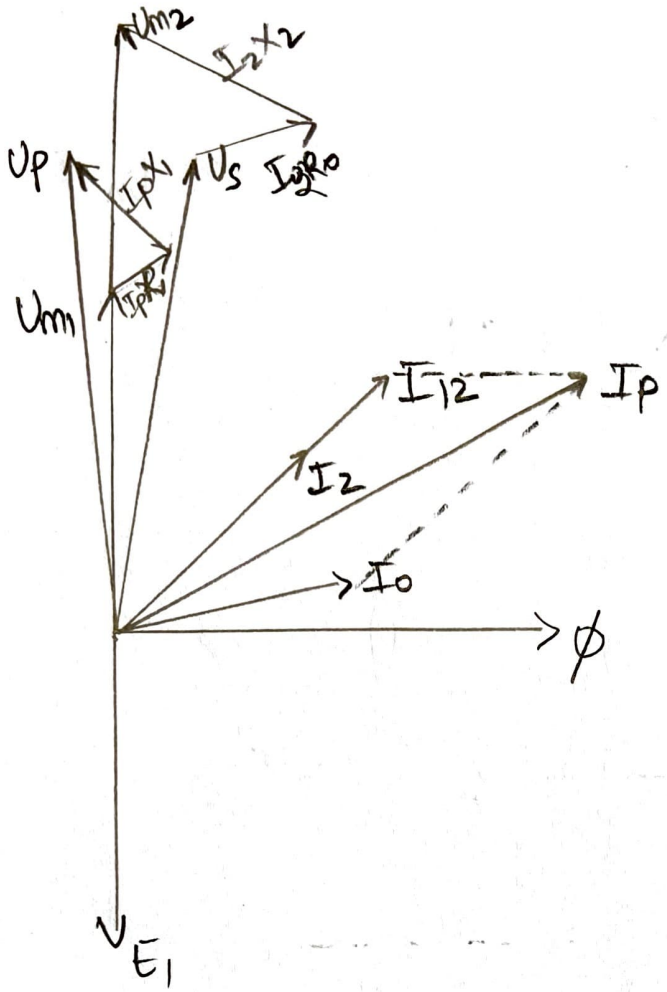






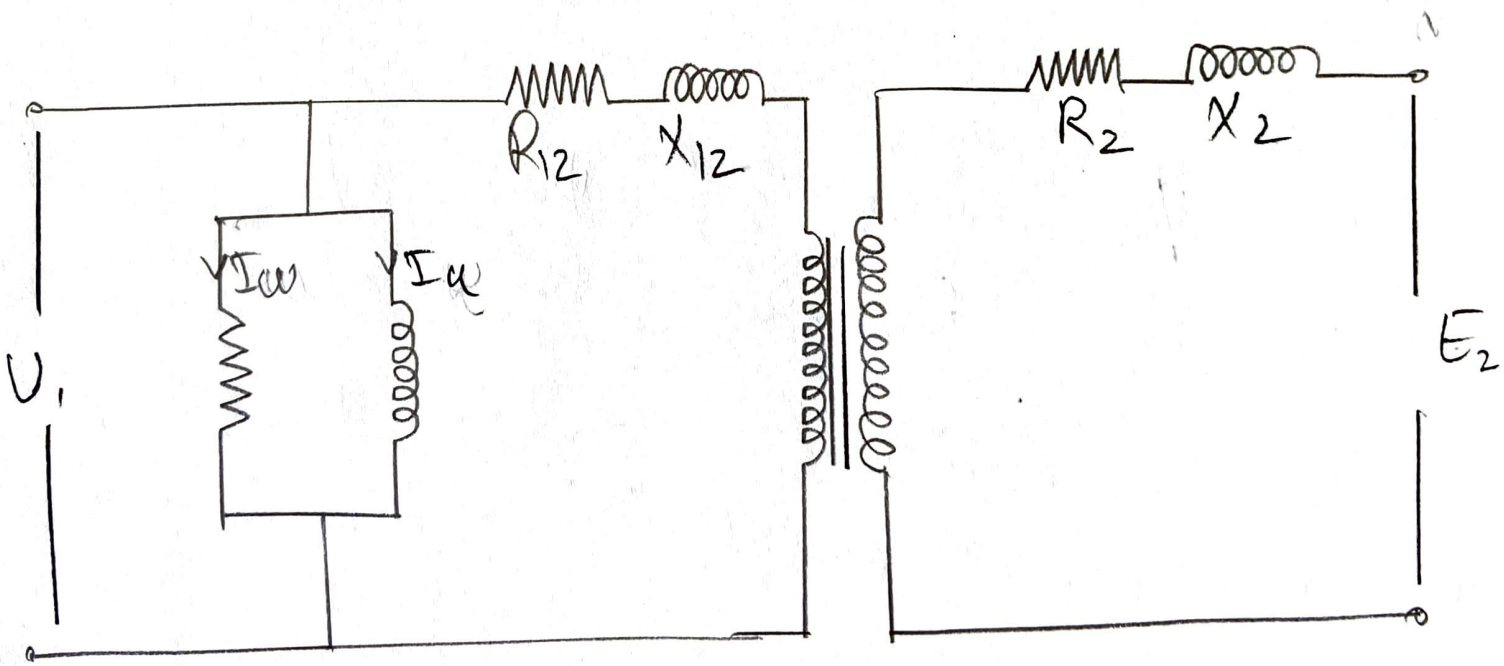
Real Transformer with Loaded Condition

$$K = \frac{V_2}{V_1} = \frac{N_2}{N_1} = \frac{I_1}{I_2}$$



$$I_{12} = \frac{I_2}{K}$$

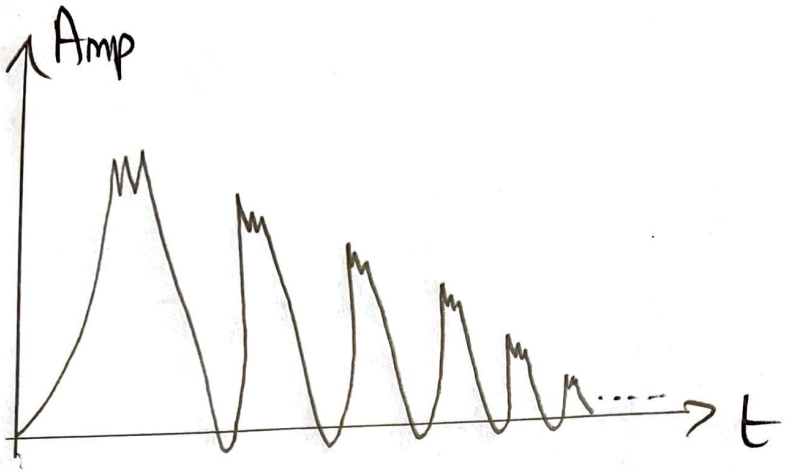
$N_1 = N_2$ } Condition
$N_1 \neq N_2$ }

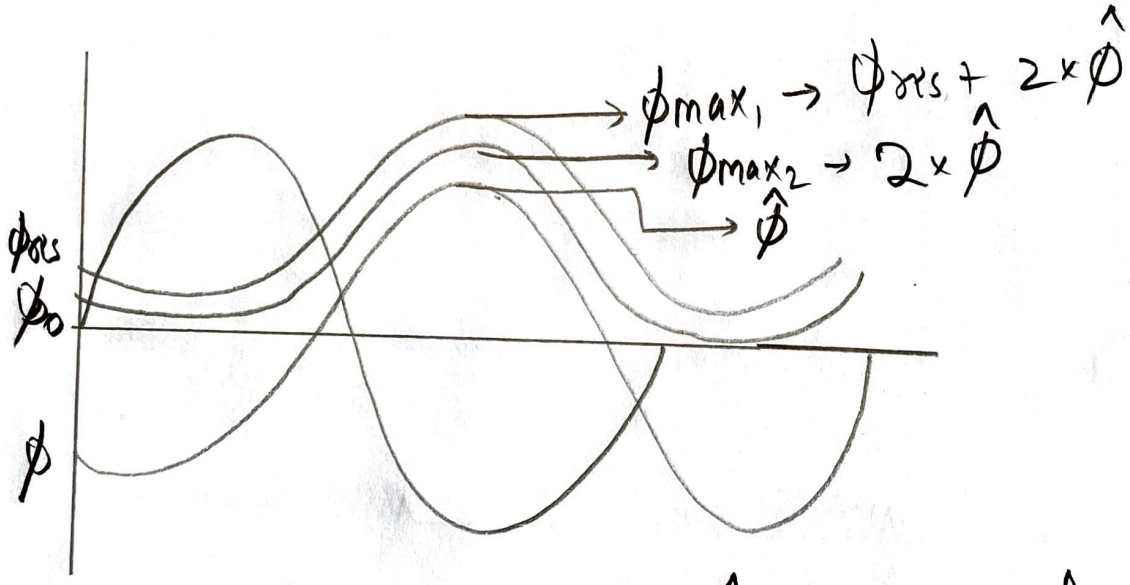


$E = 4.44 F \phi Bm A$ → Transformer design eqn

4.00 → if Square wave

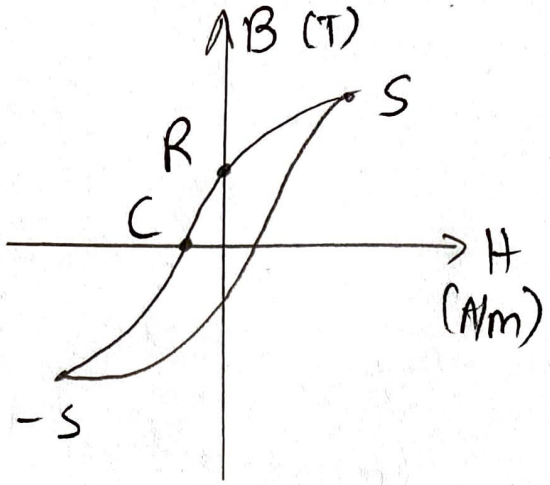
Transformer Inrush Current : (I_0)

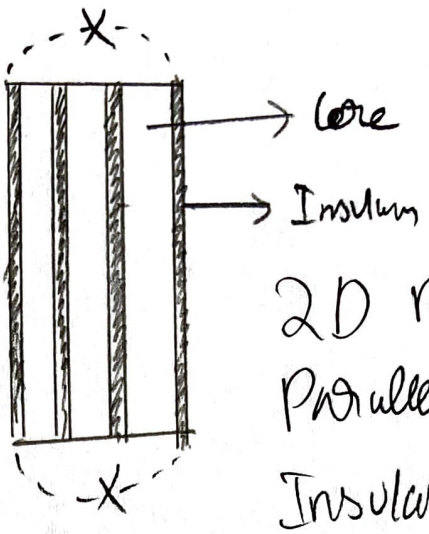




$$\phi_{max} = \phi_{res} + 2\hat{\phi} \quad ; \quad \phi_{max} = 2\hat{\phi}$$
$$B_{max} = B_{res} + 2\hat{B} \quad ; \quad B_{max} = 2\hat{B}$$

Due to Forced Sinusoidal flux 0 or ϕ_{res}, ϕ_0 or ϕ_{res} will induce DC offset Component in them.

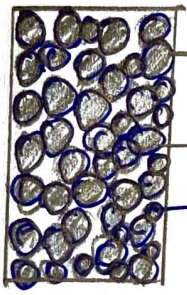




(Laminated iron core) Sheets

2D magnetism only Parallel to plates

Insulation prevent 3D Magnetism.



Powdered iron

Airgap

Insulation Coating

3D Magnetism Allows Perpendicular to Core.

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① Ideal Transformer : [no load Condition]

